

Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

Chapter 2

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Chapter Summary

- Sets (集合)
 - The Language of Sets
 - Set Operations
 - Set Identities
- Functions (函數)
 - Types of Functions
 - Operations on Functions
 - Computability
- Sequences and Summations (序列與加總)
 - Types of Sequences
 - Summation Formula
- Set Cardinality (集合基數)
 - Countable Sets
- Matrices (矩陣)
 - Matrix Arithmetic

Sets (集合)

Section 2.1

Sets

- A *set* is an unordered collection of objects.
 - the students in this class
 - the chairs in this room
- The objects in a set are called the *elements* (元素), or *members* of the set. A set is said to *contain* its elements.
- The notation $a \in A$ denotes that a is an element of the set A .
- If a is not a member of A , write $a \notin A$

Describing a Set: Roster Method (窮舉法)

- $S = \{a, b, c, d\}$
- Order not important

$$S = \{a, b, c, d\} = \{b, c, a, d\}$$

- Each distinct object is either a member or not; listing more than once does not change the set.

$$S = \{a, b, c, d\} = \{a, b, c, b, c, d\}$$

- Ellipses (...) may be used to describe a set without listing all of the members when the pattern is clear.

$$S = \{a, b, c, d, \dots, z\}$$

Roster Method

- Set of all vowels in the English alphabet:

$$V = \{a, e, i, o, u\}$$

- Set of all odd positive integers less than 10:

$$O = \{1, 3, 5, 7, 9\}$$

- Set of all positive integers less than 100:

$$S = \{1, 2, 3, \dots, 99\}$$

- Set of all integers less than 0:

$$S = \{\dots, -3, -2, -1\}$$

Some Important Sets

N = *natural numbers* (自然數) = {0, 1, 2, 3, ...}

Z = *integers* (整數) = {..., -3, -2, -1, 0, 1, 2, 3, ...}

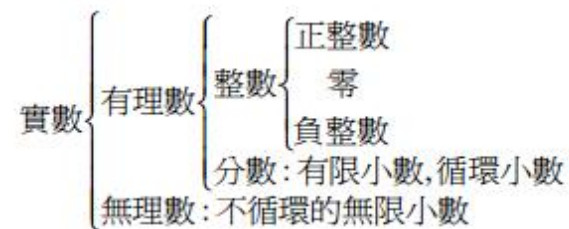
Z⁺ = *positive integers* (正整數) = {1, 2, 3, ...}

R = *set of real numbers* (實數)

R⁺ = *set of positive real numbers* (正實數)

C = *set of complex numbers* (複數)

Q = *set of rational numbers* (有理數)



Set-Builder Notation

- Specify the property or properties that all members must satisfy:

$$S = \{x \mid x \text{ is a positive integer less than } 100\}$$

$$O = \{x \mid x \text{ is an odd positive integer less than } 10\}$$

$$O = \{x \in \mathbf{Z}^+ \mid x \text{ is odd and } x < 10\}$$

- A predicate may be used:

$$S = \{x \mid P(x)\}$$

- Example: $S = \{x \mid \text{Prime}(x)\}$

- Positive rational numbers:

$$\mathbf{Q}^+ = \{x \in \mathbf{R} \mid x = p/q, \text{ for some positive integers } p, q\}$$

Interval Notation

$$[a,b] = \{x \mid a \leq x \leq b\}$$

$$[a,b) = \{x \mid a \leq x < b\}$$

$$(a,b] = \{x \mid a < x \leq b\}$$

$$(a,b) = \{x \mid a < x < b\}$$

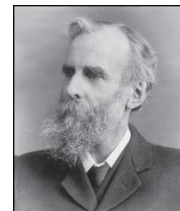
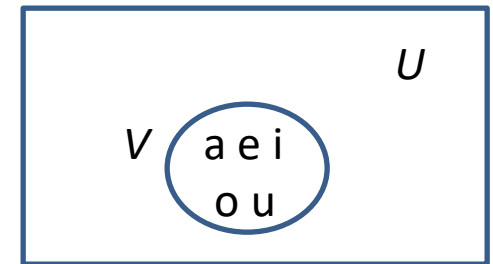
closed interval (閉區間) $[a,b]$

open interval (開區間) (a,b)

Universal Set and Empty Set

- The *universal set* (宇集合) U is the set containing everything currently under consideration.
 - Sometimes implicit
 - Sometimes explicitly stated.
 - Contents depend on the context.
- The *empty set* (空集合) is the set with no elements. Symbolized $\emptyset$, but $\{\}$ also used.

Venn Diagram



John Venn (1834-1923)
Cambridge, UK

Some things to remember

- Sets can be elements of sets.

$$\{\{1,2,3\}, a, \{b,c\}\}$$

$$\{\mathbf{N}, \mathbf{Z}, \mathbf{Q}, \mathbf{R}\}$$

- The empty set is different from a set containing the empty set.

$$\emptyset \neq \{ \emptyset \}$$

Set Equality (相等)

Definition: Two sets are *equal* if and only if they have the same elements.

- Therefore if A and B are sets, then A and B are equal if and only if $\forall x (x \in A \leftrightarrow x \in B)$
- We write $A = B$ if A and B are equal sets.

$$\{1, 3, 5\} = \{3, 5, 1\}$$

$$\{1, 5, 5, 5, 3, 3, 1\} = \{1, 3, 5\}$$

Subsets (子集合)

Definition: The set A is a *subset* of B , if and only if every element of A is also an element of B .

- The notation $A \subseteq B$ is used to indicate that A is a subset of the set B . $\forall x(x \in A \rightarrow x \in B)$
- $A \subseteq B$ holds if and only if
is true.
 1. Because $a \in \emptyset$ is always false, $\emptyset \subseteq S$, for every set S .
 2. Because $a \in S \rightarrow a \in S$, $S \subseteq S$, for every set S .

Showing a Set is or is not a Subset of Another Set

- **Showing that A is a Subset of B:** To show that $A \subseteq B$, show that if x belongs to A , then x also belongs to B .
- **Showing that A is not a Subset of B:** To show that A is not a subset of B , $A \not\subseteq B$, find an element $x \in A$ with $x \notin B$. (Such an x is a counterexample to the claim that $x \in A$ implies $x \in B$.)

Examples:

1. The set of all computer science majors at your school is a subset of all students at your school.
2. The set of integers with squares less than 100 is not a subset of the set of nonnegative integers.

Another look at Equality of Sets

- Recall that two sets A and B are *equal*, denoted by $A = B$, iff $\forall x(x \in A \leftrightarrow x \in B)$
- Using logical equivalences we have that $A = B$ iff

$$\forall x[(x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A)]$$

- This is equivalent to

$$A \subseteq B \quad \text{and} \quad B \subseteq A$$

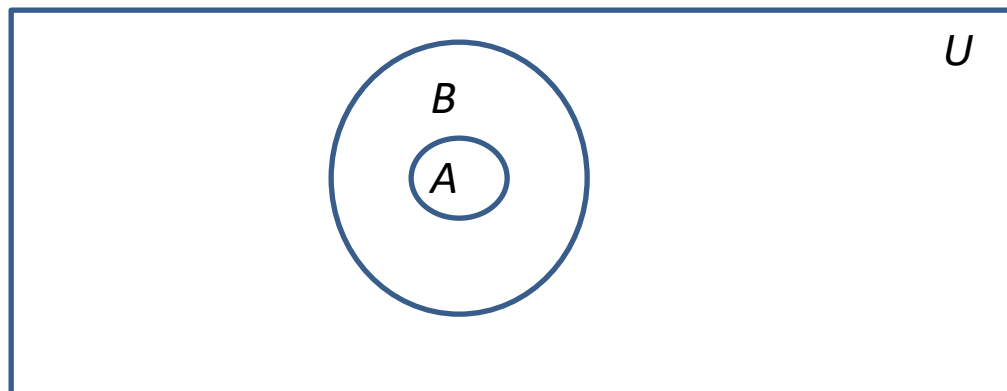
Proper Subsets (真子集)

Definition: If $A \subseteq B$, but $A \neq B$, then we say A is a proper subset of B , denoted by $A \subset B$. If $A \subset B$, then

$$\forall x(x \in A \rightarrow x \in B) \wedge \exists x(x \in B \wedge x \notin A)$$

is true.

Venn Diagram



Set Cardinality

Definition: If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is *finite*. Otherwise it is *infinite*.

Definition: The *cardinality* of a finite set A , denoted by $|A|$, is the number of (distinct) elements of A .

Examples:

1. $|\emptyset| = 0$
2. Let S be the letters of the English alphabet. Then $|S| = 26$
3. $|\{1,2,3\}| = 3$
4. $|\{\emptyset\}| = 1$
5. The set of integers is infinite.

Power Sets (幂集合)

Definition: The set of all subsets of a set A , denoted $\mathcal{P}(A)$, is called the *power set* of A .

Example: If $A = \{a, b\}$ then

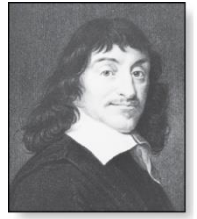
$$\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

- If a set has n elements, then the cardinality of the power set is 2^n .

Cartesian Product

- The *ordered n-tuple* (*n-元組*) $(a_1, a_2, \dots, a_n)$ is the ordered collection that has a_1 as its first element and a_2 as its second element and so on until a_n as its last element.
- Two n-tuples are equal if and only if their corresponding elements are equal.
- 2-tuples are called *ordered pairs*.
- The ordered pairs (a, b) and (c, d) are equal if and only if $a = c$ and $b = d$.

Cartesian Product



René Descartes
(1596-1650)

Definition: The *Cartesian Product* (笛卡兒乘積) of two sets A and B , denoted by $A \times B$ is the set of ordered pairs (a,b) where $a \in A$ and $b \in B$.

$$A \times B = \{(a, b) | a \in A \wedge b \in B\}$$

Example:

$$A = \{a, b\} \quad B = \{1, 2, 3\}$$

$$A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$$

Cartesian Product

Definition: The **cartesian products** of the sets $A_1, A_2, \dots, A_n$, denoted by $A_1 \times A_2 \times \dots \times A_n$, is the set of ordered n -tuples $(a_1, a_2, \dots, a_n)$ where a_i belongs to A_i , for $i = 1, \dots, n$.

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \dots, n\}$$

Example: What is $A \times B \times C$ where $A = \{0,1\}$, $B = \{1,2\}$ and $C = \{0,1,2\}$

Solution: $A \times B \times C = \{(0,1,0), (0,1,1), (0,1,2), (0,2,0), (0,2,1), (0,2,2), (1,1,0), (1,1,1), (1,1,2), (1,2,0), (1,2,1), (1,2,2)\}$

Truth Sets(真值集) and Quantifiers

- Given a predicate P and a domain D , we define the *truth set* of P to be the set of elements in D for which $P(x)$ is true. The truth set of $P(x)$ is denoted by

$$\{x \in D \mid P(x)\}$$

- Example:** The truth set of $P(x)$ where the domain is the integers and $P(x)$ is “ $|x| = 1$ ” is the set $\{-1, 1\}$

Set Operations (集合運算)

Section 2.2

Union (聯集)

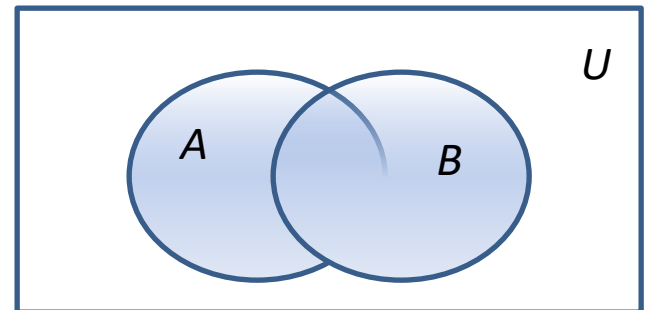
- **Definition:** Let A and B be sets. The *union* of the sets A and B , denoted by $A \cup B$, is the set:

$$\{x | x \in A \vee x \in B\}$$

- **Example:** What is $\{1,2,3\} \cup \{3,4,5\}$?

Solution: $\{1,2,3,4,5\}$

Venn Diagram for $A \cup B$



Intersection (交集)

- **Definition:** The *intersection* of sets A and B , denoted by $A \cap B$, is

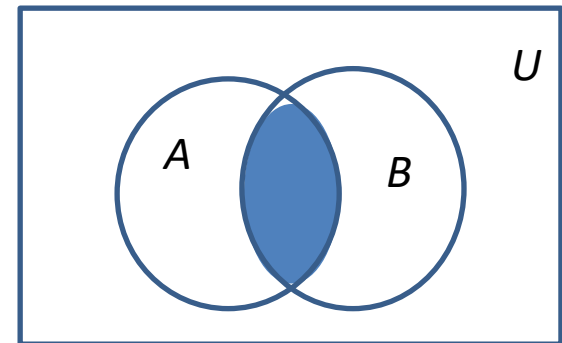
$$\{x | x \in A \wedge x \in B\}$$

- Note if the intersection is empty, then A and B are said to be *disjoint*.
- **Example:** What is? $\{1,2,3\} \cap \{3,4,5\}$?

Solution: $\{3\}$

- **Example:** What is?
 $\{1,2,3\} \cap \{4,5,6\}$?
Solution: $\emptyset$

Venn Diagram for $A \cap B$



Complement (補集)

Definition: If A is a set, then the complement of the A (with respect to U), denoted by $\bar{A}$ is the set $U - A$

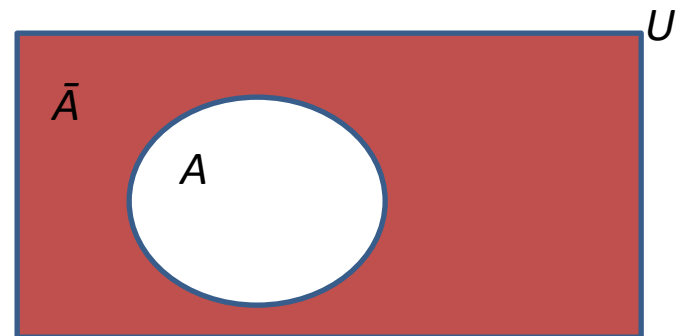
$$\bar{A} = \{x \in U \mid x \notin A\}$$

(The complement of A is sometimes denoted by A^c .)

Example: If U is the positive integers less than 100, what is the complement of $\{x \mid x > 70\}$

Solution: $\{x \mid x \leq 70\}$

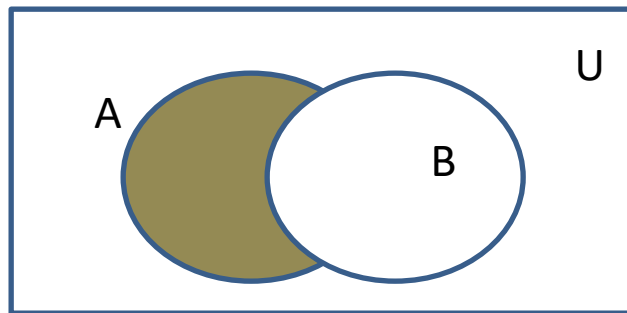
Venn Diagram for Complement



Difference (差集)

- **Definition:** Let A and B be sets. The *difference* of A and B , denoted by $A - B$, is the set containing the elements of A that are not in B . The difference of A and B is also called the complement of B with respect to A .

$$A - B = \{x \mid x \in A \wedge x \notin B\} = A \cap \bar{B}$$

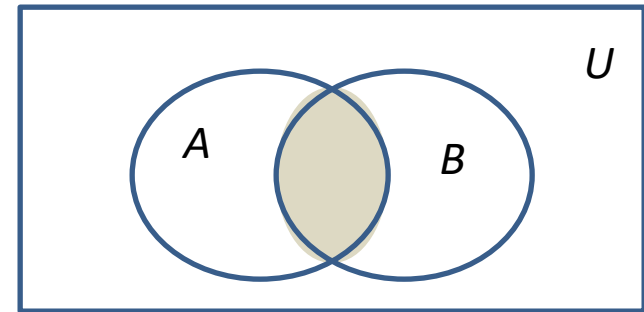


Venn Diagram for $A - B$

The Cardinality of the Union of Two Sets

- Principle of Inclusion-Exclusion (排容原理)

$$|A \cup B| = |A| + |B| - |A \cap B|$$



Venn Diagram for A , B , $A \cap B$, $A \cup B$

- **Example:** Let A be the **math majors** in your class and B be the **CS majors**. To count the number of students who are either math majors or CS majors, add the number of math majors and the number of CS majors, and subtract the number of joint CS/math majors.

Review Questions

Example: $U = \{0,1,2,3,4,5,6,7,8,9,10\}$ $A = \{1,2,3,4,5\}$, $B = \{4,5,6,7,8\}$

1. $A \cup B$

Solution: $\{1,2,3,4,5,6,7,8\}$

2. $A \cap B$

Solution: $\{4,5\}$

3. $\bar{A}$

Solution: $\{0,6,7,8,9,10\}$

4. $\bar{B}$

Solution: $\{0,1,2,3,9,10\}$

5. $A - B$

Solution: $\{1,2,3\}$

6. $B - A$

Solution: $\{6,7,8\}$

Set Identities

<i>Identity</i>	<i>Name</i>
$A \cup \emptyset = A$ $A \cap U = A$	Identity laws (同一律)
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws (支配律)
$A \cup A = A$ $A \cap A = A$	Idempotent laws (幂等律)
$\overline{\overline{A}} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws (交換律)
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Associative laws (結合律)
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	Distributive laws (分配律)
$\overline{A \cup B} = \overline{A} \cap \overline{B}$ $\overline{A \cap B} = \overline{A} \cup \overline{B}$	De Morgan's laws (笛摩根定律)
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws (吸收律)
$A \cup \overline{A} = U$ $A \cap \overline{A} = \emptyset$	Complement laws (互補律)

Proving Set Identities

- Different ways to prove set identities:
 1. Prove that each set (side of the identity) is a subset of the other.
 2. Use set builder notation and propositional logic.
 3. Membership Tables: Verify that elements in the same combination of sets always either belong or do not belong to the same side of the identity. Use 1 to indicate it is in the set and a 0 to indicate that it is not.

Proof of Second De Morgan Law

Prove that $\overline{A \cap B} = \overline{A} \cup \overline{B}$

Solution 1: We prove this identity by showing that:

1) $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$ and

$x \in \overline{A \cap B}$	by assumption
$x \notin A \cap B$	defn. of complement
$\neg((x \in A) \wedge (x \in B))$	defn. of intersection
$\neg(x \in A) \vee \neg(x \in B)$	1st De Morgan Law for Prop Logic
$x \notin A \vee x \notin B$	defn. of negation
$x \in \overline{A} \vee x \in \overline{B}$	defn. of complement
$x \in \overline{A} \cup \overline{B}$	defn. of union

2) $\overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$

$x \in \overline{A} \cup \overline{B}$	by assumption
$(x \in \overline{A}) \vee (x \in \overline{B})$	defn. of union
$(x \notin A) \vee (x \notin B)$	defn. of complement
$\neg(x \in A) \vee \neg(x \in B)$	defn. of negation
$\neg((x \in A) \wedge (x \in B))$	by 1st De Morgan Law for Prop Logic
$\neg(x \in A \cap B)$	defn. of intersection
$x \in \overline{A \cap B}$	defn. of complement

Proof of Second De Morgan Law

Prove that $\overline{A \cap B} = \overline{A} \cup \overline{B}$

Solution 2: We prove this identity using Set-Builder Notation and propositional logic:

$\overline{A \cap B}$	$=$	$\{x x \notin A \cap B\}$	by defn. of complement
	$=$	$\{x \neg(x \in (A \cap B))\}$	by defn. of does not belong symbol
	$=$	$\{x \neg(x \in A \wedge x \in B)\}$	by defn. of intersection
	$=$	$\{x \neg(x \in A) \vee \neg(x \in B)\}$	by 1st De Morgan law for Prop Logic
	$=$	$\{x x \notin A \vee x \notin B\}$	by defn. of not belong symbol
	$=$	$\{x x \in \overline{A} \vee x \in \overline{B}\}$	by defn. of complement
	$=$	$\{x x \in \overline{A} \cup \overline{B}\}$	by defn. of union
	$=$	$\overline{A} \cup \overline{B}$	by meaning of notation



Membership Table

Example: Construct a membership table to show that the distributive law holds.

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Solution:

A	B	C	$B \cap C$	$A \cup (B \cap C)$	$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$
1	1	1	1	1	1	1	1
1	1	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	0	0	0	1	1	1	1
0	1	1	1	1	1	1	1
0	1	0	0	0	1	0	0
0	0	1	0	0	0	1	0
0	0	0	0	0	0	0	0

Generalized Unions and Intersections

- Let $A_1, A_2, \dots, A_n$ be an indexed collection of sets.

We define:

$$\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup \dots \cup A_n$$

$$\bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap \dots \cap A_n$$

- Ex: For $i = 1, 2, \dots$, let $A_i = \{i, i + 1, i + 2, \dots\}$. Then,

$$\bigcup_{i=1}^n A_i = \bigcup_{i=1}^n \{i, i + 1, i + 2, \dots\} = \{1, 2, 3, \dots\}$$

$$\bigcap_{i=1}^n A_i = \bigcap_{i=1}^n \{i, i + 1, i + 2, \dots\} = \{n, n + 1, n + 2, \dots\} = A_n$$

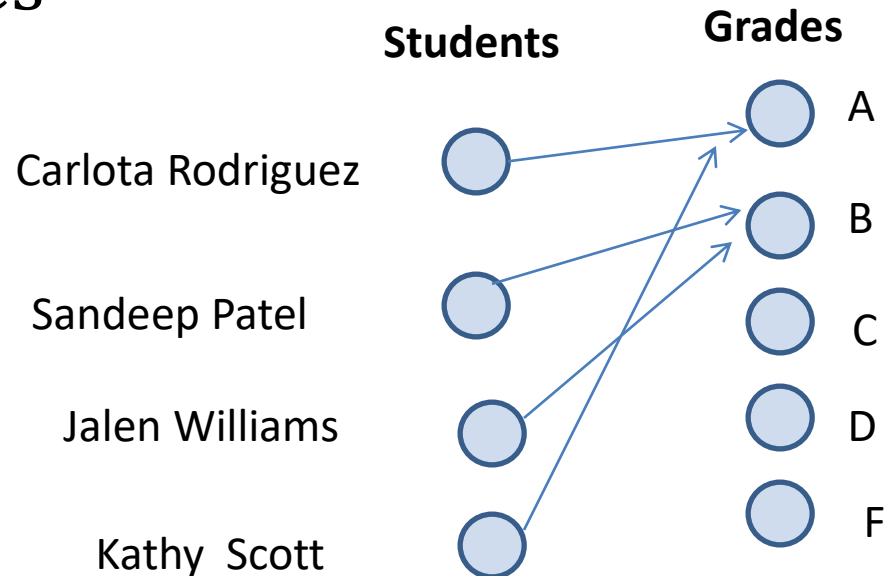
Functions(函數)

Section 2.3

Functions

Definition: Let A and B be nonempty sets. A *function* f from A to B , denoted $f: A \rightarrow B$ is an assignment of each element of A to exactly one element of B . We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A .

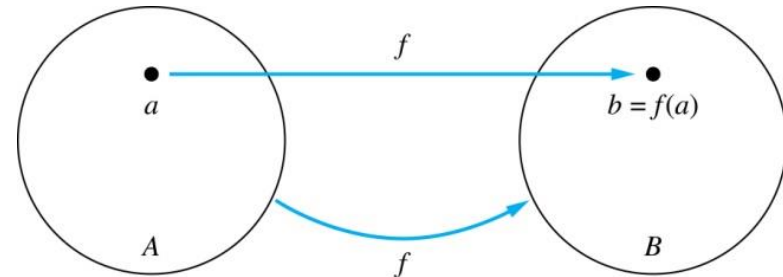
- Functions are sometimes called *mappings* or *transformations*.



Functions

Given a function $f: A \rightarrow B$:

- We say *f maps A to B* or *f is a mapping from A to B* .
- A is called the *domain* (定義域) of f .
- B is called the *codomain* (對應域) of f .
- If $f(a) = b$,
 - then b is called the *image* (映像) of a under f .
 - a is called the *preimage* (前像) of b .
- The *range* (值域) of f is the set of all images of points in A under f . We denote it by $f(A)$.
- Two functions are *equal* when they have the *same domain*, the *same codomain* and *map* each element of the *domain to the same element* of the codomain.



Representing Functions

- Functions may be specified in different ways:
 - An **explicit** statement of the **assignment**.
Students and grades example.
 - A **formula**.
$$f(x) = x + 1$$
 - A computer **program**.
 - A Java program that when given an integer n , produces the n th Fibonacci Number (covered in the next section and also in Chapter 5).

Questions

$f(a) = ?$ z

The image of d is ? z

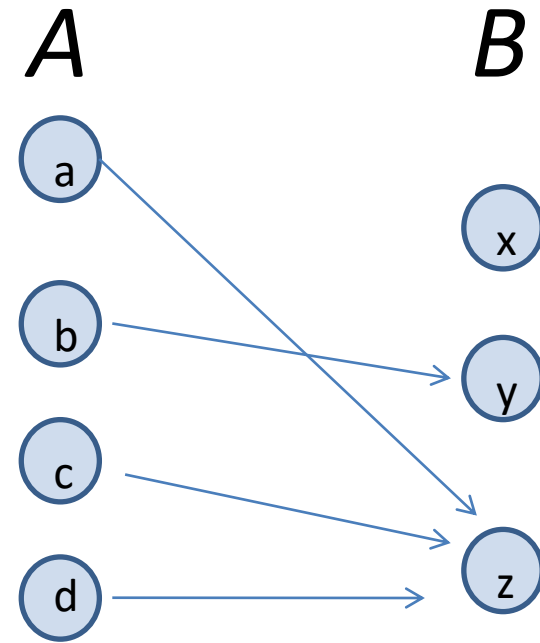
The domain of f is ? A

The codomain of f is ? B

The preimage of y is ? b

$f(A) = ?$

The preimage(s) of z is (are) ? $\{a, c, d\}$



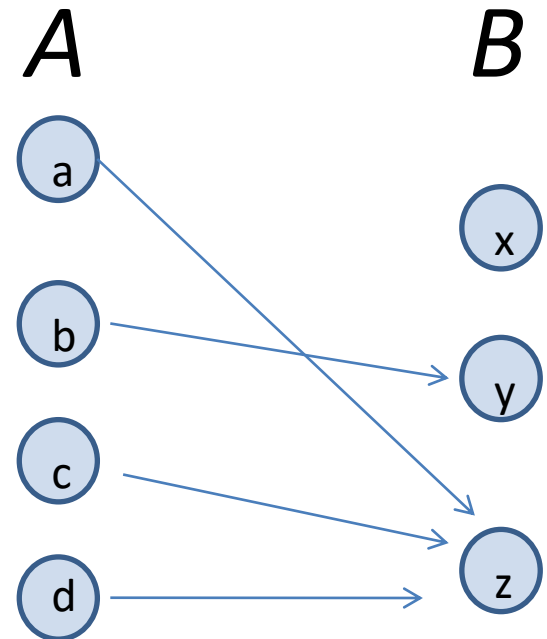
Question on Functions and Sets

- If $f : A \rightarrow B$ and S is a subset of A , then

$$f(S) = \{f(s) \mid s \in S\}$$

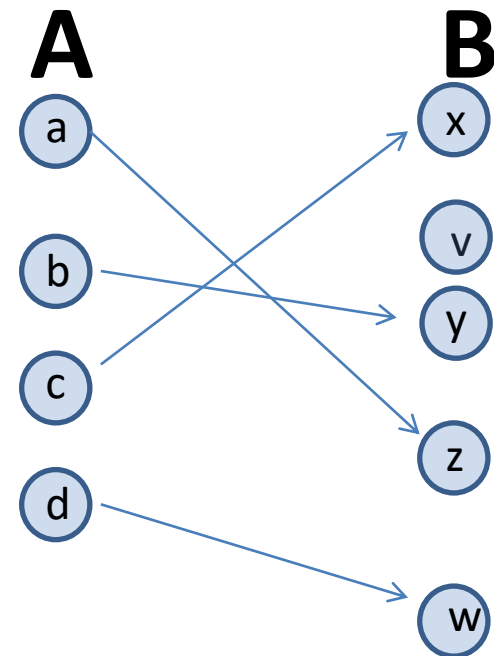
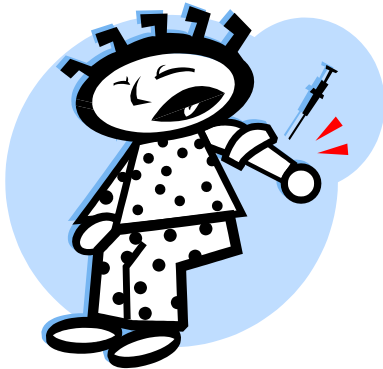
$f(\{a,b,c\})$ is ? $\{y,z\}$

$f(\{c,d\})$ is ? $\{z\}$



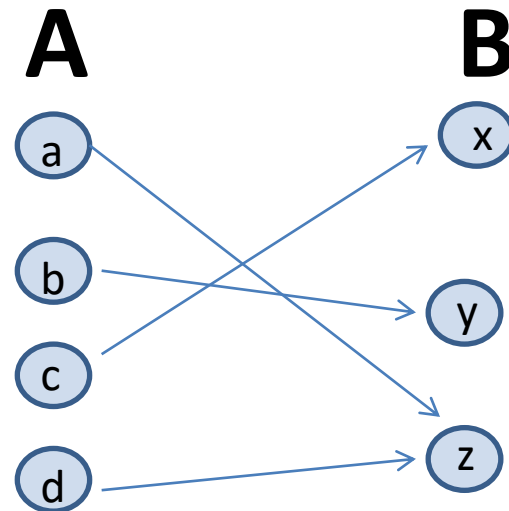
Injectons (嵌射)

Definition: A function f is said to be *one-to-one* (一對一), or *injection*, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f . A function is said to be an *injection* if it is one-to-one.



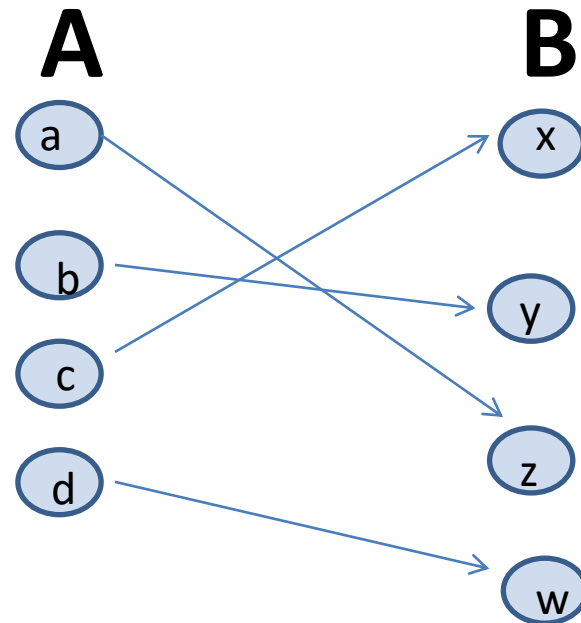
Surjections (蓋射)

Definition: A function f from A to B is called *onto* (映成) or *surjection*, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$. A function f is called a *surjection* if it is onto.



Bijections (對射)

Definition: A function f is a *one-to-one correspondence*, or a *bijection*, if it is both *one-to-one* and *onto* (surjective and injective).



Showing that f is one-to-one or onto

Suppose that $f : A \rightarrow B$.

To show that f is injective Show that if $f(x) = f(y)$ for arbitrary $x, y \in A$, then $x = y$.

To show that f is not injective Find particular elements $x, y \in A$ such that $x \neq y$ and $f(x) = f(y)$.

To show that f is surjective Consider an arbitrary element $y \in B$ and find an element $x \in A$ such that $f(x) = y$.

To show that f is not surjective Find a particular $y \in B$ such that $f(x) \neq y$ for all $x \in A$.

Showing that f is one-to-one or onto

Example 1: Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3\}$ defined by $f(a) = 3$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$. Is f an onto function?

Solution: Yes, f is onto since all three elements of the codomain are images of elements in the domain. If the codomain were changed to $\{1, 2, 3, 4\}$, f would not be onto.

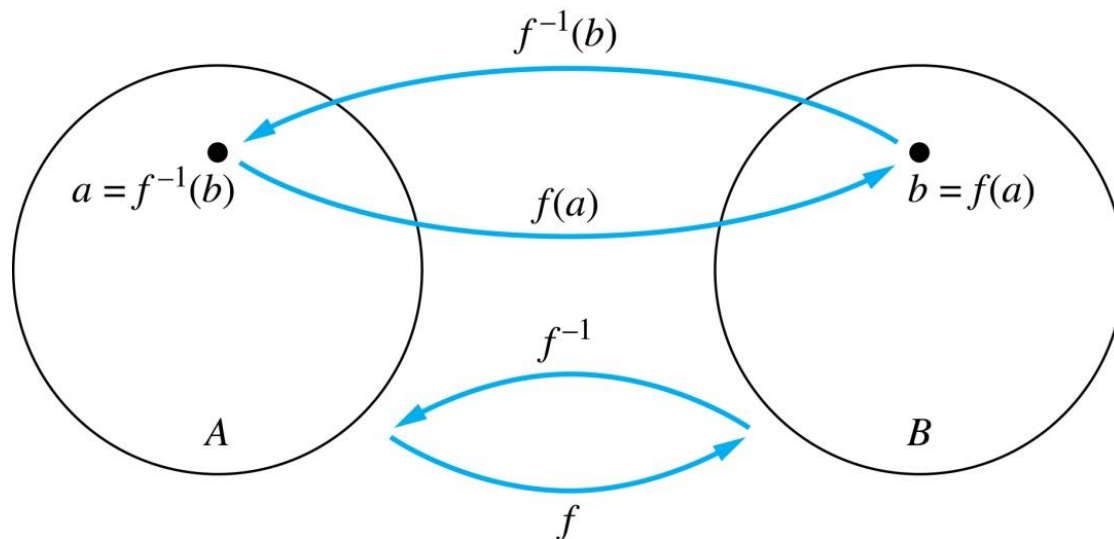
Example 2: Is the function $f(x) = x^2$ from the set of integers onto?

Solution: No, f is not onto because there is no integer x with $x^2 = -1$, for example.

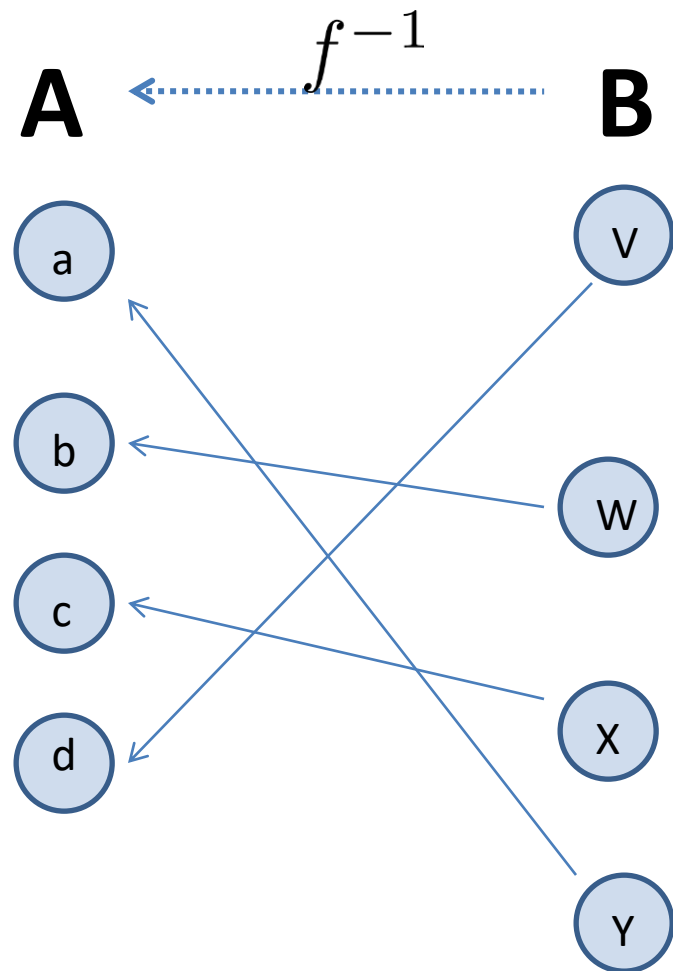
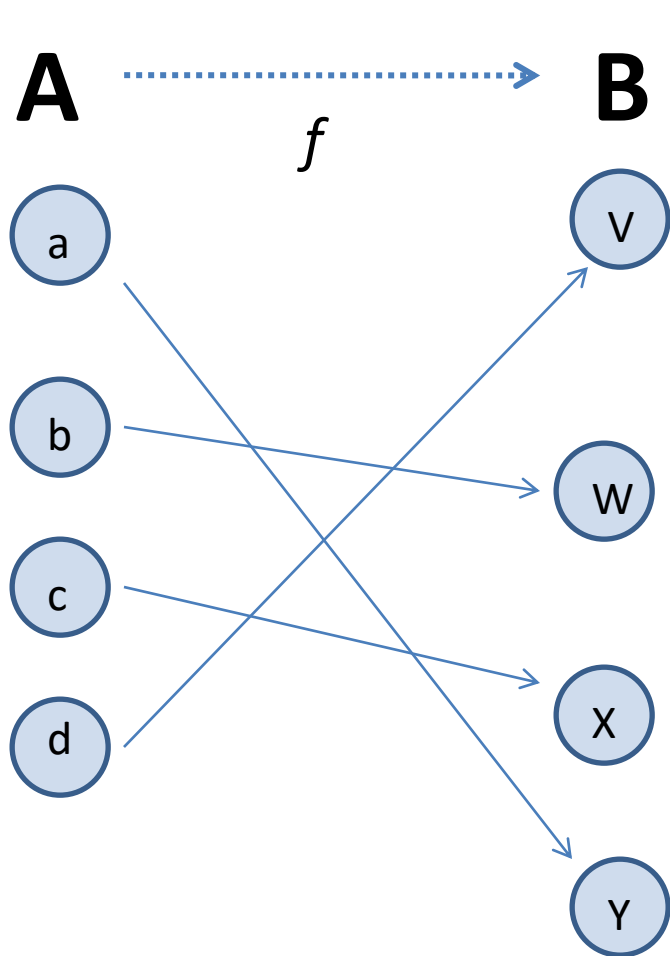
Inverse Functions (反函數)

Definition: Let f be a bijection from A to B . Then the *inverse* of f , denoted f^{-1} , is the function from B to A defined as $f^{-1}(y) = x$ iff $f(x) = y$

No inverse exists unless f is a bijection. Why?



Inverse Functions



Questions

Example 1: Let f be the function from $\{a,b,c\}$ to $\{1,2,3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Is f invertible and if so what is its inverse?

Solution: The function f is invertible because it is a **one-to-one correspondence**. The inverse function f^{-1} reverses the correspondence given by f , so $f^{-1}(1) = c$, $f^{-1}(2) = a$, and $f^{-1}(3) = b$.

Questions

Example 2: Let $f: \mathbf{Z} \rightarrow \mathbf{Z}$ be such that $f(x) = x + 1$.
Is f invertible, and if so, what is its inverse?

Solution: The function f is invertible because it is a **one-to-one correspondence**. The inverse function f^{-1} reverses the correspondence so $f^{-1}(y) = y - 1$.

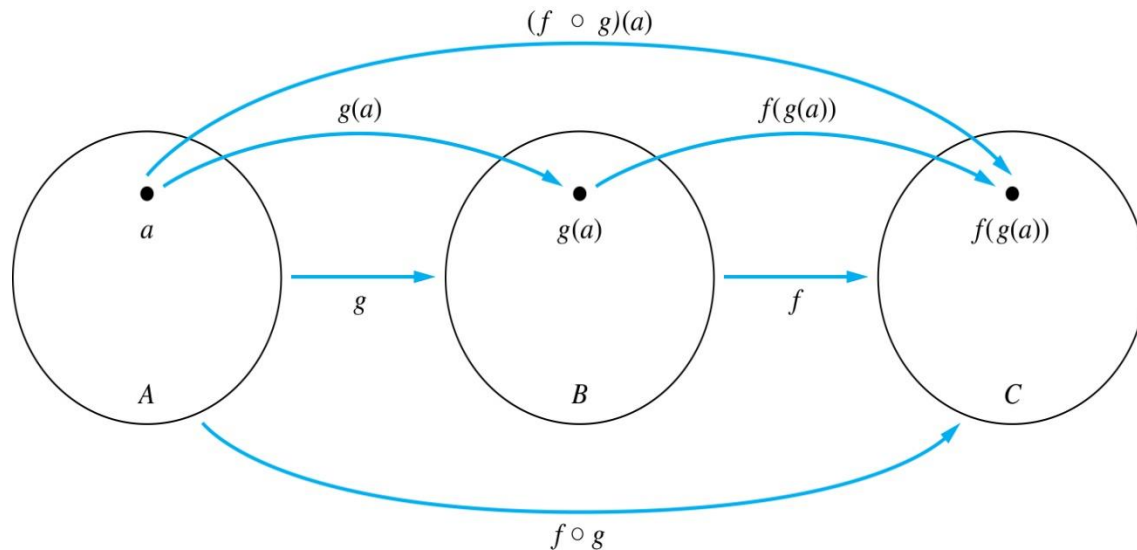
Questions

Example 3: Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be such that $f(x) = x^2$.
Is f invertible, and if so, what is its inverse?

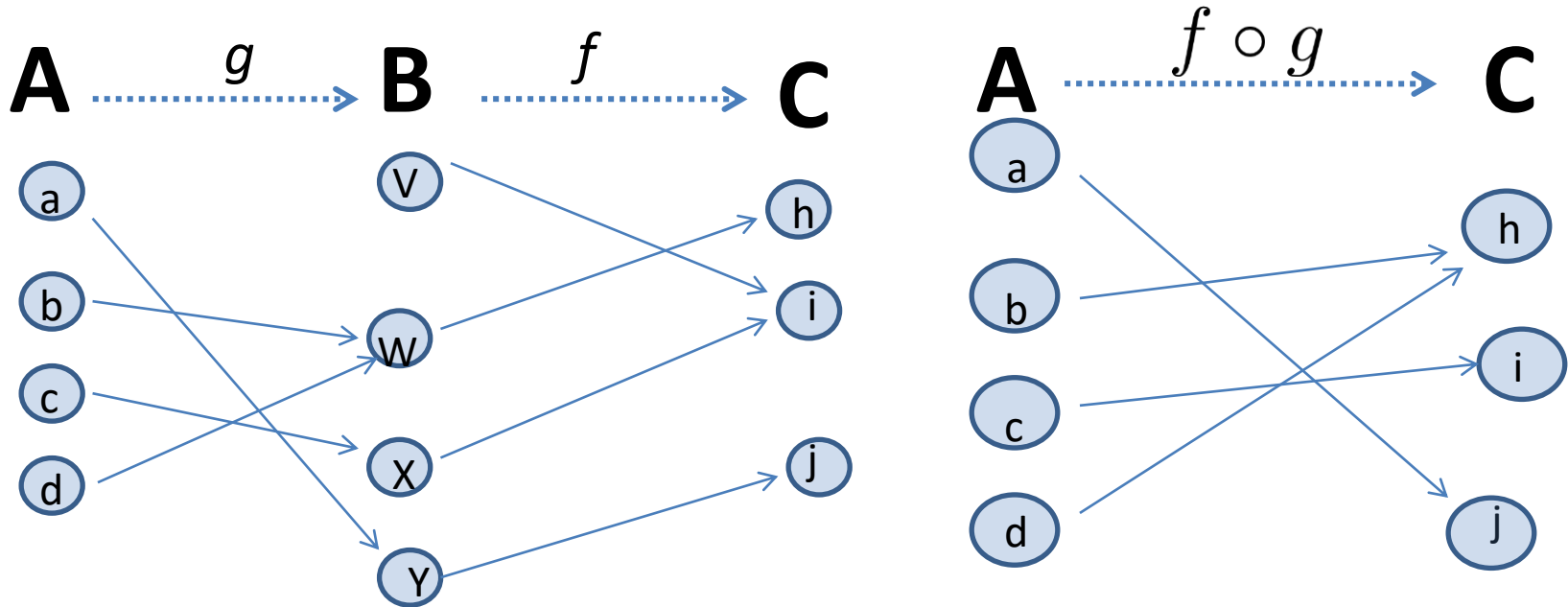
Solution: The function f is not invertible because it is **not one-to-one**.

Composition (合成函數)

- **Definition:** Let $f: B \rightarrow C$, $g: A \rightarrow B$. The *composition of f with g* , denoted $f \circ g$ is the function from A to C defined by $f \circ g(x) = f(g(x))$



Composition



Composition

Example 1: If $f(x) = x^2$ and $g(x) = 2x + 1$, then

$$f(g(x)) = (2x + 1)^2$$

and

$$g(f(x)) = 2x^2 + 1$$

Composition Questions

Example 2: Let g be the function from the set $\{a,b,c\}$ to itself such that $g(a) = b$, $g(b) = c$, and $g(c) = a$. Let f be the function from the set $\{a,b,c\}$ to the set $\{1,2,3\}$ such that $f(a) = 3$, $f(b) = 2$, and $f(c) = 1$.

What is the composition of f with g , and what is the composition of g with f .

Solution: The composition $f \circ g$ is defined by

$$f \circ g (a) = f(g(a)) = f(b) = 2.$$

$$f \circ g (b) = f(g(b)) = f(c) = 1.$$

$$f \circ g (c) = f(g(c)) = f(a) = 3.$$

Note that $g \circ f$ is not defined, because the range of f is not a subset of the domain of g .

Composition Questions

Example 2: Let f and g be functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$.

What is the **composition of f with g** , and also the composition of g with f ?

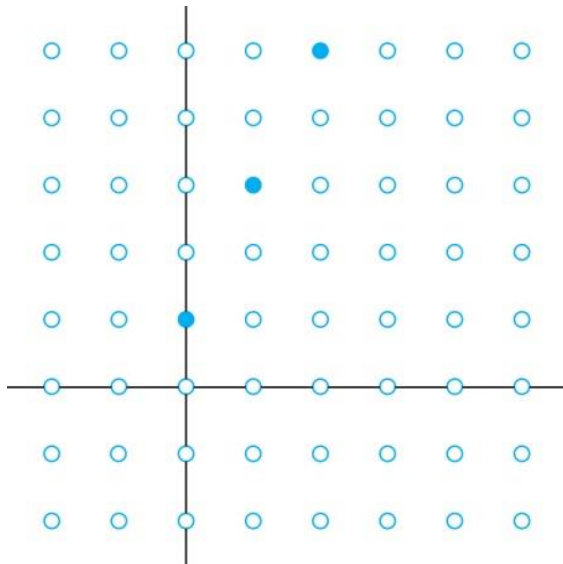
Solution:

$$f \circ g (x) = f(g(x)) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 7$$

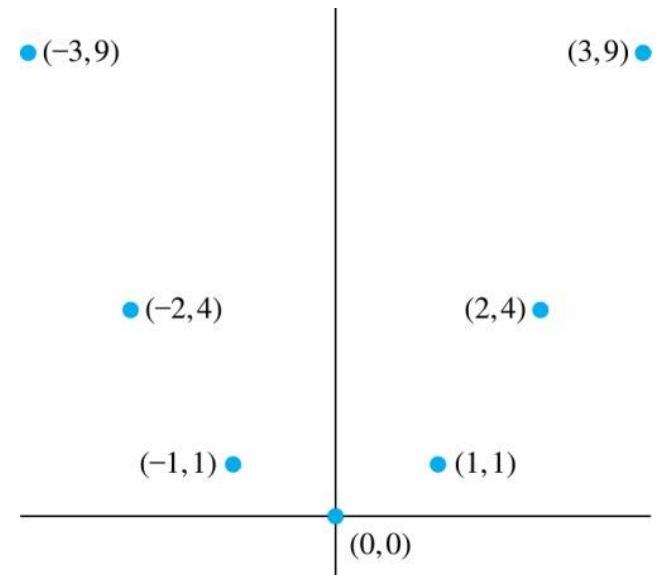
$$g \circ f (x) = g(f(x)) = g(2x + 3) = 3(2x + 3) + 2 = 6x + 11$$

Graphs of Functions

- Let f be a function from the set A to the set B . The *graph of the function f* is the set of ordered pairs $\{(a,b) \mid a \in A \text{ and } f(a) = b\}$.



Graph of $f(n) = 2n + 1$ from $\mathbb{Z}$ to $\mathbb{Z}$



Graph of $f(x) = x^2$ from $\mathbb{Z}$ to $\mathbb{Z}$

Some Important Functions

- The *floor* function(底函數), denoted $f(x) = \lfloor x \rfloor$ is the largest integer less than or equal to x .
- The *ceiling* function(頂函數), denoted $f(x) = \lceil x \rceil$ is the smallest integer greater than or equal to x

Example:

$$\lceil 3.5 \rceil = 4$$

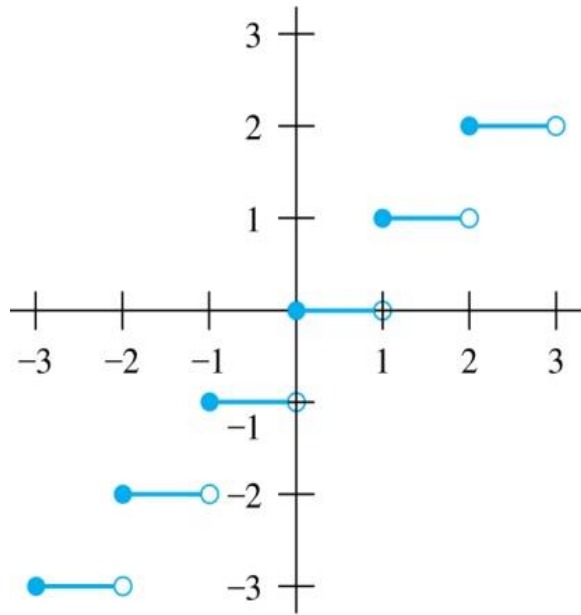
$$\lfloor 3.5 \rfloor = 3$$

$$\lceil -1.5 \rceil = -1$$

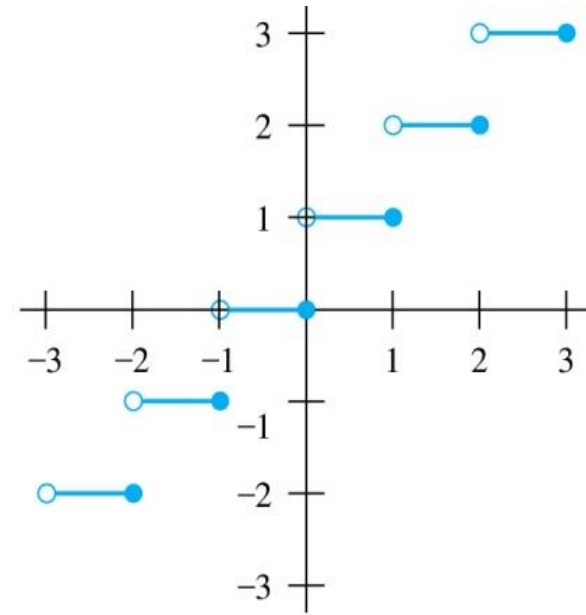
$$\lfloor -1.5 \rfloor = -2$$

Floor and Ceiling Functions

(Reading Exercise)



(a) $y = \lfloor x \rfloor$



(b) $y = \lceil x \rceil$

Graph of (a) Floor and (b) Ceiling Functions

Floor and Ceiling Functions

TABLE 1 Useful Properties of the Floor and Ceiling Functions.

(n is an integer, x is a real number)

(1a) $\lfloor x \rfloor = n$ if and only if $n \leq x < n + 1$

(1b) $\lceil x \rceil = n$ if and only if $n - 1 < x \leq n$

(1c) $\lfloor x \rfloor = n$ if and only if $x - 1 < n \leq x$

(1d) $\lceil x \rceil = n$ if and only if $x \leq n < x + 1$

(2) $x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1$

(3a) $\lfloor -x \rfloor = -\lceil x \rceil$

(3b) $\lceil -x \rceil = -\lfloor x \rfloor$

(4a) $\lfloor x + n \rfloor = \lfloor x \rfloor + n$

(4b) $\lceil x + n \rceil = \lceil x \rceil + n$

Proving Properties of Functions

(Reading Exercise for Detail)

Example: Prove that x is a real number, then

$$\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + 1/2 \rfloor$$

Solution: Let $x = n + \varepsilon$, where n is an integer and $0 \leq \varepsilon < 1$.

Case 1: $\varepsilon < 1/2$

- $2x = 2n + 2\varepsilon$ and $\lfloor 2x \rfloor = 2n$, since $0 \leq 2\varepsilon < 1$.
- $\lfloor x + 1/2 \rfloor = n$, since $x + 1/2 = n + (1/2 + \varepsilon)$ and $0 \leq 1/2 + \varepsilon < 1$.
- Hence, $\lfloor 2x \rfloor = 2n$ and $\lfloor x \rfloor + \lfloor x + 1/2 \rfloor = n + n = 2n$.

Case 2: $\varepsilon \geq 1/2$

- $2x = 2n + 2\varepsilon = (2n + 1) + (2\varepsilon - 1)$ and $\lfloor 2x \rfloor = 2n + 1$, since $0 \leq 2\varepsilon - 1 < 1$.
- $\lfloor x + 1/2 \rfloor = \lfloor n + (1/2 + \varepsilon) \rfloor = \lfloor n + 1 + (\varepsilon - 1/2) \rfloor = n + 1$ since $0 \leq \varepsilon - 1/2 < 1$.
- Hence, $\lfloor 2x \rfloor = 2n + 1$ and $\lfloor x \rfloor + \lfloor x + 1/2 \rfloor = n + (n + 1) = 2n + 1$.



Factorial Function (階層函數)

Definition: $f: \mathbf{N} \rightarrow \mathbf{Z}^+$, denoted by $f(n) = n!$ is the product of the first n positive integers when n is a nonnegative integer.

$$f(n) = 1 \cdot 2 \cdots (n-1) \cdot n, \quad f(0) = 0! = 1$$

Examples:

$$f(1) = 1! = 1$$

$$f(2) = 2! = 1 \cdot 2 = 2$$

$$f(6) = 6! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 = 720$$

$$f(20) = 2,432,902,008,176,640,000.$$

$$\text{Stirling's Formula: } n! \sim \sqrt{2\pi n} (n/e)^n$$

$$(f(n) \sim g(n) \doteq \lim_{n \rightarrow \infty} f(n)/g(n) = 1)$$

Sequences(序列) and Summations(加總)

Section 2.4

Sequences

- Sequences are **ordered lists of elements**.
 - 1, 2, 3, 5, 8
 - 1, 3, 9, 27, 81,

Definition: A **sequence** is a function **from** a subset of the integers (usually either the set $\{0, 1, 2, 3, 4, \dots\}$ or $\{1, 2, 3, 4, \dots\}$) **to** a set S .

- The notation a_n is used to denote the image of the integer n . We can think of a_n as the **equivalent of $f(n)$** where f is a function from $\{0, 1, 2, \dots\}$ to S . We call a_n a **term** of the sequence.

Sequences

Example: Consider the sequence $\{a_n\}$ where

$$a_n = \frac{1}{n} \quad \{a_n\} = \{a_1, a_2, a_3, \dots\}$$

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4} \dots$$

Geometric Progression (幾何/等比數列)

Definition: A *geometric progression* is a sequence of the form: $a, ar, ar^2, \dots, ar^n, \dots$

where the *initial term* (首項) a and the *common ratio* (公比) r are real numbers.

Examples:

1. Let $a = 1$ and $r = -1$. Then:

$$\{b_n\} = \{b_0, b_1, b_2, b_3, b_4, \dots\} = \{1, -1, 1, -1, 1, \dots\}$$

2. Let $a = 2$ and $r = 5$. Then:

$$\{c_n\} = \{c_0, c_1, c_2, c_3, c_4, \dots\} = \{2, 10, 50, 250, 1250, \dots\}$$

3. Let $a = 6$ and $r = 1/3$. Then:

$$\{d_n\} = \{d_0, d_1, d_2, d_3, d_4, \dots\} = \{6, 2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \dots\}$$

Arithmetic Progression(算數/等差數列)

Definition: A *arithmetic progression* is a sequence of the form:
 $a, a + d, a + 2d, \dots, a + nd, \dots$
where the *initial term*(首項) a and the *common difference*(公差) d are real numbers.

Examples:

1. Let $a = -1$ and $d = 4$:

$$\{s_n\} = \{s_0, s_1, s_2, s_3, s_4, \dots\} = \{-1, 3, 7, 11, 15, \dots\}$$

2. Let $a = 7$ and $d = -3$:

$$\{t_n\} = \{t_0, t_1, t_2, t_3, t_4, \dots\} = \{7, 4, 1, -2, -5, \dots\}$$

3. Let $a = 1$ and $d = 2$:

$$\{u_n\} = \{u_0, u_1, u_2, u_3, u_4, \dots\} = \{1, 3, 5, 7, 9, \dots\}$$

Strings

Definition: A *string* is a finite sequence of characters from a finite set (an alphabet).

- Sequences of characters or bits are important in computer science.
- The *empty string* is represented by λ .
- The string *abcde* has *length* 5.

Recurrence Relations(遞迴關係)

Definition: A *recurrence relation* for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence, namely, $a_0, a_1, \dots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer.

- A sequence is called a *solution of a recurrence relation* if its terms satisfy the recurrence relation.
- The *initial conditions* for a sequence specify the terms that precede the first term where the recurrence relation takes effect.

Questions about Recurrence Relations

Example 1: Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 1, 2, 3, 4, \dots$ and suppose that $a_0 = 2$. What are a_1 , a_2 and a_3 ?
[Here $a_0 = 2$ is the initial condition.]

Solution: We see from the recurrence relation that

$$a_1 = a_0 + 3 = 2 + 3 = 5$$

$$a_2 = 5 + 3 = 8$$

$$a_3 = 8 + 3 = 11$$

Questions about Recurrence Relations

Example 2: Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} - a_{n-2}$ for $n = 2, 3, 4, \dots$ and suppose that $a_0 = 3$ and $a_1 = 5$. What are a_2 and a_3 ?
[Here the initial conditions are $a_0 = 3$ and $a_1 = 5$.]

Solution: We see from the recurrence relation that

$$a_2 = a_1 - a_0 = 5 - 3 = 2$$

$$a_3 = a_2 - a_1 = 2 - 5 = -3$$

Fibonacci Sequence(費伯納西數列)

Definition: Define the *Fibonacci sequence*, $f_0, f_1, f_2, \dots$, by:

- Initial Conditions: $f_0 = 0, f_1 = 1$
- Recurrence Relation: $f_n = f_{n-1} + f_{n-2}$

Example: Find f_2, f_3, f_4, f_5 and f_6 .

Answer:

$$f_2 = f_1 + f_0 = 1 + 0 = 1,$$

$$f_3 = f_2 + f_1 = 1 + 1 = 2,$$

$$f_4 = f_3 + f_2 = 2 + 1 = 3,$$

$$f_5 = f_4 + f_3 = 3 + 2 = 5,$$

$$f_6 = f_5 + f_4 = 5 + 3 = 8.$$

Solving Recurrence Relations

- Finding a formula for the n th term of the sequence generated by a recurrence relation is called *solving the recurrence relation*.
- Such a formula is called a *closed formula*.
- Various methods for solving recurrence relations will be covered in Chapter 8 where recurrence relations will be studied in greater depth.
- Here we illustrate by example the method of iteration in which we need to guess the formula. The guess can be proved correct by the method of induction (Chapter 5).

Iterative Solution Example

Method 1: Working upward, forward substitution

Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 2, 3, 4, \dots$ and suppose that $a_1 = 2$.

$$a_2 = 2 + 3$$

$$a_3 = a_2 + 3 = (2 + 3) + 3 = 2 + 3 \cdot 2$$

$$a_4 = a_3 + 3 = (2 + 2 \cdot 3) + 3 = 2 + 3 \cdot 3$$

.

.

.

$$a_n = a_{n-1} + 3 = (2 + 3 \cdot (n - 2)) + 3 = 2 + 3(n - 1)$$

Iterative Solution Example

Method 2: Working downward, **backward substitution**

Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 2, 3, 4, \dots$ and suppose that $a_1 = 2$.

$$\begin{aligned}a_n &= a_{n-1} + 3 \\&= (a_{n-2} + 3) + 3 = a_{n-2} + 3 \cdot 2 \\&= (a_{n-3} + 3) + 3 \cdot 2 = a_{n-3} + 3 \cdot 3\end{aligned}$$

.

.

.

$$\begin{aligned}&= a_2 + 3(n-2) = (a_1 + 3) + 3(n-2) = a_1 + 3(n-1) \\&= 2 + 3(n-1)\end{aligned}$$

Financial Application

(Reading Exercise)

Example: Suppose that a person deposits (存款) \$10,000.00 in a savings account at a bank yielding 11% per year with interest compounded annually. How much will be in the account **after 30 years?**

Let P_n denote the amount in the account after 30 years. P_n satisfies the following recurrence relation:

$$P_n = P_{n-1} + 0.11P_{n-1} = (1.11) P_{n-1}$$

with the initial condition $P_0 = 10,000$

Continued on next slide →

Financial Application

(Reading Exercise)

$$P_n = P_{n-1} + 0.11P_{n-1} = (1.11) P_{n-1}$$

with the initial condition $P_0 = 10,000$

Solution: Forward Substitution

$$P_1 = (1.11)P_0$$

$$P_2 = (1.11)P_1 = (1.11)^2 P_0$$

$$P_3 = (1.11)P_2 = (1.11)^3 P_0$$

:

$$P_n = (1.11)P_{n-1} = (1.11)^n P_0 = (1.11)^n 10,000$$

$$P_n = (1.11)^n 10,000 \text{ (Can prove by induction, covered in Chapter 5)}$$

$$P_{30} = (1.11)^{30} 10,000 = \$228,992.97$$

Questions on Special Integer Sequences (*opt*)

Example 1: Find formulae for the sequences with the following first five terms: 1, $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$

Solution: Note that the denominators are powers of 2. The sequence with $a_n = 1/2^n$ is a possible match. This is a geometric progression with $a = 1$ and $r = \frac{1}{2}$.

Example 2: Consider 1,3,5,7,9

Solution: Note that each term is obtained by adding 2 to the previous term. A possible formula is $a_n = 2n + 1$. This is an arithmetic progression with $a = 1$ and $d = 2$.

Example 3: 1, -1, 1, -1,1

Solution: The terms alternate between 1 and -1. A possible sequence is $a_n = (-1)^n$. This is a geometric progression with $a = 1$ and $r = -1$.

Useful Sequences

TABLE 1 Some Useful Sequences.

<i>n</i> th Term	First 10 Terms
n^2	1, 4, 9, 16, 25, 36, 49, 64, 81, 100, ...
n^3	1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, ...
n^4	1, 16, 81, 256, 625, 1296, 2401, 4096, 6561, 10000, ...
2^n	2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, ...
3^n	3, 9, 27, 81, 243, 729, 2187, 6561, 19683, 59049, ...
$n!$	1, 2, 6, 24, 120, 720, 5040, 40320, 362880, 3628800, ...
f_n	1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, ...

Interesting Integer Sequences

(Work out with fun!)

- Here are three interesting sequences to try from the OEIS site. To solve each puzzle, find a rule that determines the terms of the sequence.
- Guess the rules for forming for the following sequences:
 - 2, 3, 3, 5, 10, 13, 39, 43, 172, 177, ...
 - Hint: Think of adding and multiplying by numbers to generate this sequence.
 - 0, 0, 0, 0, 4, 9, 5, 1, 1, 0, 55, ...
 - Hint: Think of the English names for the numbers representing the position in the sequence and the Roman Numerals for the same number.
 - 2, 4, 6, 30, 32, 34, 36, 40, 42, 44, 46, ...
 - Hint: Think of the English names for numbers, and whether or **not** they have the letter 'e.'
- The answers and many more can be found at <http://oeis.org/Spuzzle.html>

Summations

- Sum of the terms $a_m, a_{m+1}, \dots, a_n$ from the sequence $\{a_n\}$
- The notation:

$$\sum_{j=m}^n a_j \quad \sum_{j=m}^n a_j \quad \sum_{m \leq j \leq n} a_j$$

represents

$$a_m + a_{m+1} + \dots + a_n$$

- The variable j is called the *index of summation*. It runs through all the integers starting with its *lower limit* m and ending with its *upper limit* n .

Summations

- More generally for a set S :

$$\sum_{j \in S} a_j$$

- **Examples:** $r^0 + r^1 + r^2 + r^3 + \dots + r^n = \sum_0^n r^j$

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots = \sum_1^{\infty} \frac{1}{i}$$

If $S = \{2, 5, 7, 10\}$ then $\sum_{j \in S} a_j = a_2 + a_5 + a_7 + a_{10}$

Product Notation (*optional*)

- Product of the terms $a_m, a_{m+1}, \dots, a_n$
from the sequence $\{a_n\}$

- The notation:

$$\prod_{j=m}^n a_j \qquad \prod_{j=m}^n a_j \qquad \prod_{m \leq j \leq n} a_j$$

represents $a_m \times a_{m+1} \times \dots \times a_n$

Geometric Series

Sums of terms of geometric progressions

$$\sum_{j=0}^n ar^j = \begin{cases} \frac{ar^{n+1} - a}{r - 1} & r \neq 1 \\ (n + 1)a & r = 1 \end{cases}$$

Proof: Let $S_n = \sum_{j=0}^n ar^j$

$$rS_n = r \sum_{j=0}^n ar^j = \sum_{j=0}^n ar^{j+1}$$

To compute S_n , first multiply both sides of the equality by r and then manipulate the resulting sum

$$= \sum_{k=1}^{n+1} ar^k$$

Shifting the index of summation with $k = j + 1$.

$$= \left(\sum_{k=0}^n ar^k \right) + (ar^{n+1} - a)$$

Removing $k = n + 1$ term and adding $k = 0$ term.

$$= S_n + (ar^{n+1} - a)$$

Substituting S for summation formula

$$\therefore rS_n = S_n + (ar^{n+1} - a)$$

$$S_n = \frac{ar^{n+1} - a}{r - 1} \text{ if } r \neq 1; \quad \text{and} \quad S_n = \sum_{j=0}^n ar^j = \sum_{j=0}^n a = (n + 1)a \text{ if } r = 1$$

Some Useful Summation Formulae

TABLE 2 Some Useful Summation Formulae.	
<i>Sum</i>	<i>Closed Form</i>
$\sum_{k=0}^n ar^k \ (r \neq 0)$	$\frac{ar^{n+1} - a}{r - 1}, r \neq 1$
$\sum_{k=1}^n k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^n k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^n k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, x < 1$	$\frac{1}{1-x}$
$\sum_{k=1}^{\infty} kx^{k-1}, x < 1$	$\frac{1}{(1-x)^2}$

Geometric Series: We just proved this.

Later we will prove some of these **by induction**.

Proof in text (pp. 176-177) (requires calculus)

Cardinality of Sets

Section 2.5

Cardinality

Definition: The *cardinality* (基數) of a set A is equal to the cardinality of a set B , denoted

$$|A| = |B|,$$

if and only if there is a *one-to-one correspondence (i.e., a bijection)* from A to B

- If there is a one-to-one function (i.e., an injection) from A to B , the cardinality of A is less than or the same as the cardinality of B and we write $|A| \leq |B|$.
- When $|A| \leq |B|$ and A and B have different cardinality, we say that the cardinality of A is less than the cardinality of B and write $|A| < |B|$.

Cardinality

- **Definition:** A set that is either finite or has the same cardinality as the set of positive integers ($\mathbf{Z}^+$) is called *countable* (可數). A set that is not countable is *uncountable* (不可數).
- The set of real numbers $\mathbf{R}$ is an uncountable set.
- When an infinite set is countable (*countably infinite*) its cardinality is $\aleph_0$ (where $\aleph$ is aleph, the 1st letter of the Hebrew (希伯來) alphabet). We write $|S| = \aleph_0$ and say that S has cardinality “aleph null.”

Showing that a Set is Countable

- An infinite set is countable if and only if it is possible to list the elements of the set in a sequence (indexed by the positive integers).
- The reason for this is that a one-to-one correspondence f from the set of positive integers to a set S can be expressed in terms of a sequence $a_1, a_2, \dots, a_n, \dots$ where $a_1 = f(1)$, $a_2 = f(2), \dots, a_n = f(n), \dots$

Hilbert's Grand Hotel

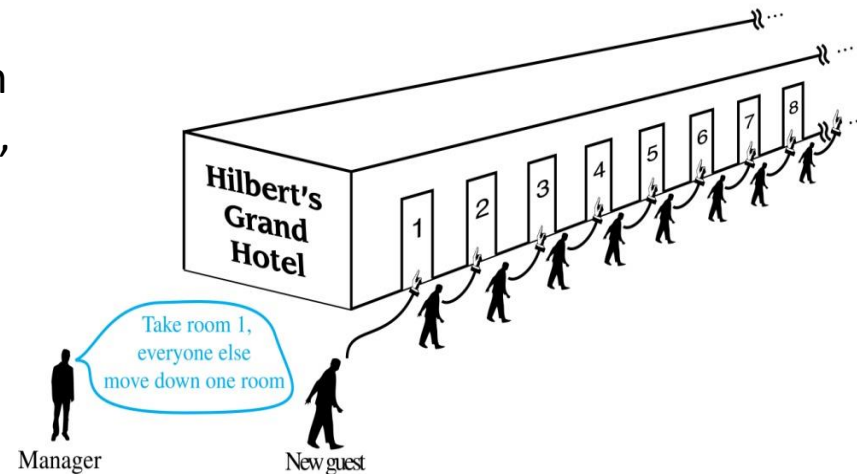
(Have fun!)



David Hilbert

The Grand Hotel (example due to David Hilbert) has countably infinite number of rooms. We can always accommodate a new guest arriving at the fully occupied Grand Hotel. How is this possible?

Explanation: Because the rooms of Grand Hotel are countable, we can list them as Room 1, Room 2, Room 3, and so on. When a new guest arrives, we move the guest in Room 1 to Room 2, the guest in Room 2 to Room 3, and in general the guest in Room n to Room $n + 1$, for all positive integers n . This frees up Room 1, which we assign to the new guest, and all the current guests still have rooms.

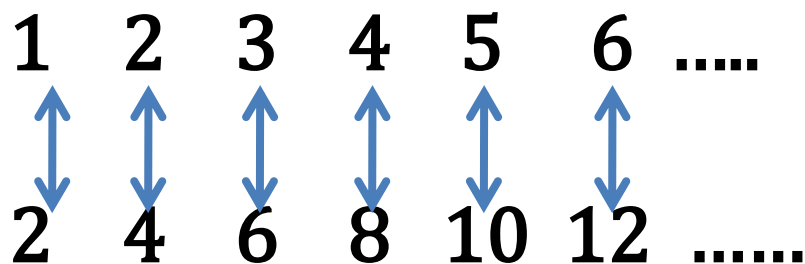


<https://www.youtube.com/watch?v=TEisKmlSEv0>

Showing that a Set is Countable

Example 1: Show that the set of positive **even** integers E is countable set.

Solution: Let $f(x) = 2x$.



Then f is a bijection from $\mathbf{Z}^+$ to E since f is both one-to-one and onto. To show that it is one-to-one, suppose that $f(n) = f(m)$. Then $2n = 2m$, and so $n = m$. To see that it is onto, suppose that t is an even positive integer. Then $t = 2k$ for some positive integer k and $f(k) = t$. ◀

Showing that a Set is Countable

Example 2: Show that the set of integers $\mathbf{Z}$ is countable.

Solution: Can list in a sequence:

0, 1, -1 , 2, -2 , 3, -3 ,

Or can define a bijection from $\mathbf{Z}^+$ to $\mathbf{Z}$:

- When n is even: $f(n) = n/2$
- When n is odd: $f(n) = -(n-1)/2$



The Positive Rational Numbers (有理數) are Countable

- **Definition:** A *rational number* (有理數) can be expressed as the ratio of two integers p and q where $q \neq 0$.
 - $\frac{3}{4}$ is a rational number
 - $\sqrt{2}$ is not a rational number.

Example 3: Show that the positive rational numbers are countable.

Solution: The positive rational numbers are countable since they can be arranged in a sequence:

$$r_1, r_2, r_3, \dots$$

The next slide shows how this is done.

→

The Positive Rational Numbers are Countable

First row $q = 1$.

Second row $q = 2$.

etc.

Constructing the List

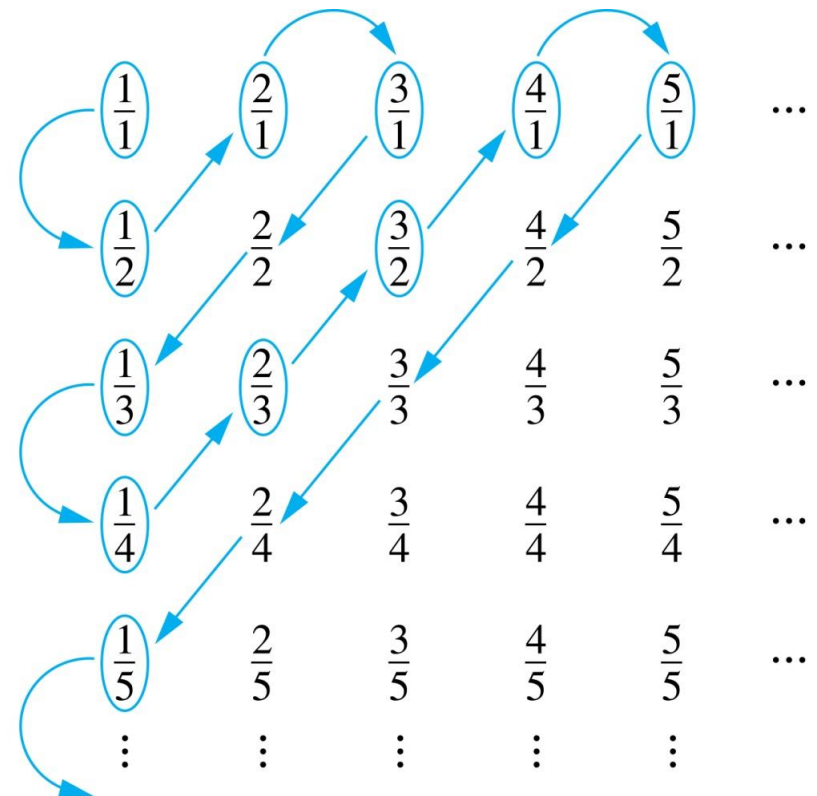
First list p/q with $p + q = 2$.

Next list p/q with $p + q = 3$

And so on.

$1, \frac{1}{2}, 2, 3, \frac{1}{3}, \frac{1}{4}, \frac{2}{3}, \dots$

Terms not circled
are not listed
because they
repeat previously
listed terms



Strings

(Reading Exercise)

Example 4: Show that the set of finite strings S over a finite alphabet A is countably infinite.

Assume an alphabetical ordering of symbols in A

Solution: Show that the strings can be listed in a sequence. First list

1. All the strings of length 0 in alphabetical order.
2. Then all the strings of length 1 in lexicographic (as in a dictionary) order.
3. Then all the strings of length 2 in lexicographic order.
4. And so on.

This implies a bijection from $\mathbf{N}$ to S and hence it is a countably infinite set.



The set of all Java programs is countable. (Reading Exercise)

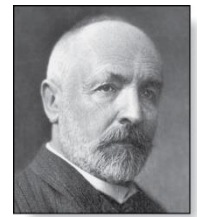
Example 5: Show that the set of all Java programs is countable.

Solution: Let S be the set of strings constructed from the characters which can appear in a Java program. Use the ordering from the previous example. Take each string in turn:

- Feed the string into a Java compiler. (A Java compiler will determine if the input program is a syntactically correct Java program.)
- If the compiler says YES, this is a syntactically correct Java program, we add the program to the list.
- We move on to the next string.

In this way we construct an implied bijection from $\mathbf{N}$ to the set of Java programs. Hence, the set of Java programs is countable. ◀

The Real Numbers are Uncountable



Georg Cantor
(1845-1918)

Example: Show that the set of real numbers is uncountable.

Solution: The method is called the **Cantor diagonalization argument**, and is a **proof by contradiction**.

1. Suppose $\mathbf{R}$ is countable. Then the real numbers between 0 and 1 are also countable (any subset of a countable set is countable - an exercise in the text).
2. The real numbers between 0 and 1 can be listed in order $r_1, r_2, r_3, \dots$.
3. Let the decimal representation of this listing be
$$\begin{aligned}r_1 &= 0.d_{11}d_{12}d_{13}d_{14}d_{15}d_{16}\dots \\r_2 &= 0.d_{21}d_{22}d_{23}d_{24}d_{25}d_{26}\dots \\r_3 &= 0.d_{31}d_{32}d_{33}d_{34}d_{35}d_{36}\dots \\&\vdots\end{aligned}$$
4. Form **a new real number** with the decimal expansion $r = .d_1d_2d_3d_4\dots$
where $d_i = 3$ if $d_{ii} \neq 3$ and $d_i = 4$ if $d_{ii} = 3$.
5. r is not equal to any of the $r_1, r_2, r_3, \dots$ Because it differs from r_i in its i th position after the decimal point. Therefore there is a real number between 0 and 1 that is not on the list since every real number has a unique decimal expansion. Hence, all the real numbers between 0 and 1 cannot be listed, so the set of real numbers between 0 and 1 is uncountable.
6. Since a set with an uncountable subset is uncountable (an exercise), the set of real numbers is uncountable. ◀

Matrices (矩陣)

Section 2.6

Matrices

- Matrices are useful discrete structures that can be used in many ways. For example, they are used to:
 - describe certain types of functions known as **linear transformations**.
 - Express which vertices of a **graph** are connected by edges (see Chapter 10).
- In later chapters, we will see matrices used to build models of:
 - Transportation systems.
 - Communication networks.
- Algorithms based on matrix models will be presented in later chapters.
- Here we cover the matrix arithmetic that will be needed later.

Matrix

Definition: A *matrix* is a rectangular array of numbers. A matrix with m rows and n columns is called an $m \times n$ matrix.

- The plural(複數) of matrix is *matrices*.
- A matrix with the same number of rows as columns is called *square*.
- Two matrices are *equal* if they have the same number of rows and the same number of columns and the corresponding entries in every position are equal.

3×2 matrix

$$\begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 3 \end{bmatrix}$$

Notation

- Let m and n be positive integers and let

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

- The i th row of $\mathbf{A}$ is the $1 \times n$ matrix $[a_{i1}, a_{i2}, \dots, a_{in}]$.
The j th column of $\mathbf{A}$ is the $m \times 1$ matrix:

$$\begin{bmatrix} a_{1j} \\ a_{2j} \\ \cdot \\ \cdot \\ a_{mj} \end{bmatrix}$$

- The (i,j) th *element* or *entry* of $\mathbf{A}$ is the element a_{ij} . We can use $\mathbf{A} = [a_{ij}]$ to denote the matrix with its (i,j) th element equal to a_{ij} .

Matrix Arithmetic: Addition

Defintion: Let $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$ be $m \times n$ matrices. The sum of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{A} + \mathbf{B}$, is the $m \times n$ matrix that has $a_{ij} + b_{ij}$ as its (i,j) th element. In other words, $\mathbf{A} + \mathbf{B} = [a_{ij} + b_{ij}]$.

Example:

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 2 & -3 \\ 3 & 4 & 0 \end{bmatrix} + \begin{bmatrix} 3 & 4 & -1 \\ 1 & -3 & 0 \\ -1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 4 & -2 \\ 3 & -1 & -3 \\ 2 & 5 & 2 \end{bmatrix}$$

Note that matrices of different sizes can not be added.

Matrix Multiplication

Definition: Let $\mathbf{A}$ be an $m \times k$ matrix and $\mathbf{B}$ be a $k \times n$ matrix. The *product* of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{AB}$, is the $m \times n$ matrix that has its (i,j) th element equal to the sum of the products of the corresponding elements from the *i th row of $\mathbf{A}$* and the *j th column of $\mathbf{B}$* . In other words, if $\mathbf{AB} = [c_{ij}]$ then $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ik}b_{kj}$.

Example:

$$\begin{bmatrix} 1 & 0 & 4 \\ 2 & 1 & 1 \\ 3 & 1 & 0 \\ 0 & 2 & 2 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 1 & 1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 14 & 4 \\ 8 & 9 \\ 7 & 13 \\ 8 & 2 \end{bmatrix}$$

The product of two matrices is undefined when the number of columns in the first matrix is not the same as the number of rows in the second.

Illustration of Matrix Multiplication

- The Product of $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \vdots & \vdots & & \vdots \\ \vdots & \vdots & & \vdots \\ \textcolor{red}{a_{i1}} & \textcolor{red}{a_{i2}} & \dots & \textcolor{red}{a_{ik}} \\ \vdots & \vdots & & \vdots \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mk} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & \dots & \textcolor{red}{b_{1j}} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & \textcolor{red}{b_{2j}} & \dots & b_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ \vdots & \vdots & & \vdots & & \vdots \\ b_{k1} & b_{k2} & \dots & \textcolor{red}{b_{kj}} & \dots & b_{kn} \end{bmatrix}$$
$$\mathbf{AB} = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \vdots & \vdots & & \vdots \\ \vdots & \vdots & \textcolor{red}{c_{ij}} & \vdots \\ \vdots & \vdots & & \vdots \\ c_{m1} & c_{m2} & \dots & c_{mn} \end{bmatrix}$$

$$\textcolor{red}{c_{ij}} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ik}b_{kj}$$

Matrix Multiplication is **not** Commutative

Example: Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

Does **$\mathbf{AB} = \mathbf{BA}$** ?

Solution:

$$\mathbf{AB} = \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix}$$

$$\mathbf{BA} = \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}$$

$$\mathbf{AB} \neq \mathbf{BA}$$

Identity Matrix (單位矩陣) and Powers (幕次) of Matrices

Definition: The *identity matrix of order n* is the $n \times n$ matrix $\mathbf{I}_n = [\delta_{ij}]$, where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$.

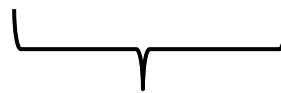
$$\mathbf{I}_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & & \cdot \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

$$\mathbf{A}\mathbf{I}_n = \mathbf{I}_m\mathbf{A} = \mathbf{A}$$

when $\mathbf{A}$ is an $m \times n$ matrix

Powers of **square matrices** can be defined. When $\mathbf{A}$ is an $n \times n$ matrix, we have:

$$\mathbf{A}^0 = \mathbf{I}_n \quad \mathbf{A}^r = \underbrace{\mathbf{A}\mathbf{A}\mathbf{A}\cdots\mathbf{A}}_{r \text{ times}}$$


 r times

Transposes (轉置) of Matrices

Definition: Let $\mathbf{A} = [a_{ij}]$ be an $m \times n$ matrix. The *transpose* of $\mathbf{A}$, denoted by $\mathbf{A}^t$, is the $n \times m$ matrix obtained by interchanging the rows and columns of $\mathbf{A}$.

If $\mathbf{A}^t = [b_{ij}]$, then $b_{ij} = a_{ji}$ for $i = 1, 2, \dots, n$
and $j = 1, 2, \dots, m$.

The transpose of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ is the matrix $\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$.

Transposes of Matrices

Definition: A square matrix $\mathbf{A}$ is called **symmetric(對稱)** if $\mathbf{A} = \mathbf{A}^t$. Thus $\mathbf{A} = [a_{ij}]$ is symmetric if $a_{ij} = a_{ji}$ for i and j with $1 \leq i \leq n$ and $1 \leq j \leq n$.

The matrix $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ is symmetric

Zero-One Matrices (零壹矩陣)

Definition: A matrix all of whose **entries are either 0 or 1** is called a ***zero-one matrix***.

(These will be used in Chapters 9 and 10.)

Algorithms operating on discrete structures represented by zero-one matrices are based on Boolean arithmetic defined by the following Boolean operations:

$$b_1 \wedge b_2 = \begin{cases} 1 & \text{if } b_1 = b_2 = 1 \\ 0 & \text{otherwise} \end{cases} \quad b_1 \vee b_2 = \begin{cases} 1 & \text{if } b_1 = 1 \text{ or } b_2 = 1 \\ 0 & \text{otherwise} \end{cases}$$

Zero-One Matrices

Definition: Let $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$ be an $m \times n$ zero-one matrices.

- The *join* of $\mathbf{A}$ and $\mathbf{B}$ is the zero-one matrix with (i,j) th entry $a_{ij} \vee b_{ij}$. The *join* of $\mathbf{A}$ and $\mathbf{B}$ is denoted by $\mathbf{A} \vee \mathbf{B}$.
- The *meet* of $\mathbf{A}$ and $\mathbf{B}$ is the zero-one matrix with (i,j) th entry $a_{ij} \wedge b_{ij}$. The *meet* of $\mathbf{A}$ and $\mathbf{B}$ is denoted by $\mathbf{A} \wedge \mathbf{B}$.

Joins and Meets of Zero-One Matrices

Example: Find the join and meet of the zero-one matrices

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Solution: The join of **A** and **B** is

$$\mathbf{A} \vee \mathbf{B} = \begin{bmatrix} 1 \vee 0 & 0 \vee 1 & 1 \vee 0 \\ 0 \vee 1 & 1 \vee 1 & 0 \vee 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

The meet of **A** and **B** is

$$\mathbf{A} \wedge \mathbf{B} = \begin{bmatrix} 1 \wedge 0 & 0 \wedge 1 & 1 \wedge 0 \\ 0 \wedge 1 & 1 \wedge 1 & 0 \wedge 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Boolean Product of Zero-One Matrices

Definition: Let $\mathbf{A} = [a_{ij}]$ be an $m \times k$ zero-one matrix and $\mathbf{B} = [b_{ij}]$ be a $k \times n$ zero-one matrix. The *Boolean product* of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{A} \odot \mathbf{B}$, is the $m \times n$ zero-one matrix with (i,j) th entry

$$c_{ij} = (a_{i1} \wedge b_{1j}) \vee (a_{i2} \wedge b_{2j}) \vee \dots \vee (a_{ik} \wedge b_{kj}).$$

Example: Find the Boolean product of $\mathbf{A}$ and $\mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Solution: The Boolean product $\mathbf{A} \odot \mathbf{B}$ is given by

$$\begin{aligned} \mathbf{A} \odot \mathbf{B} &= \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \\ (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 1) \vee (1 \wedge 1) & (0 \wedge 0) \vee (1 \wedge 1) \\ (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \end{bmatrix} \\ &= \begin{bmatrix} 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \\ 0 \vee 0 & 0 \vee 1 & 0 \vee 1 \\ 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}. \end{aligned}$$

Boolean Powers of Zero-One Matrices

Definition: Let $\mathbf{A}$ be a **square zero-one matrix** and let r be a positive integer. The r th **Boolean power** of $\mathbf{A}$ is the Boolean product of r factors of $\mathbf{A}$, denoted by $\mathbf{A}^{[r]}$. Hence,

$$\mathbf{A}^{[r]} = \underbrace{\mathbf{A} \odot \mathbf{A} \odot \dots \odot \mathbf{A}}_{r \text{ times}}.$$

We define $\mathbf{A}^{[0]}$ to be $\mathbf{I}_n$.

Boolean Powers of Zero-One Matrices

Example: Let

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Find $\mathbf{A}^{[n]}$ for all positive integers n .

Solution:

$$\mathbf{A}^{[2]} = \mathbf{A} \odot \mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad \mathbf{A}^{[3]} = \mathbf{A}^{[2]} \odot \mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\mathbf{A}^{[4]} = \mathbf{A}^{[3]} \odot \mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\mathbf{A}^{[5]} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad \mathbf{A}^{[n]} = \mathbf{A}^{[5]} \quad \text{for all positive integers } n \text{ with } n \geq 5.$$